

3-b Family Facts in Prolog

PROGRAM:

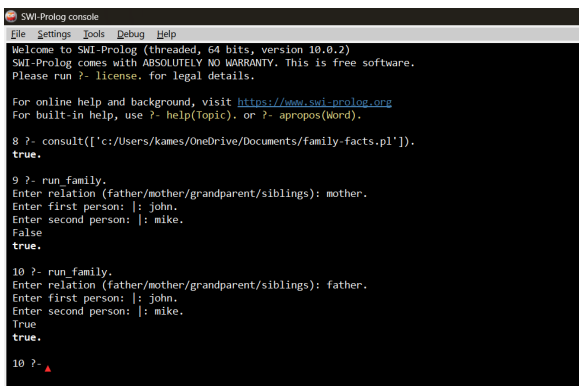
```
% Facts
parent(john, mike).
parent(john, susan).
parent(mary, mike).
parent(mary, susan).

male(john).
male(mike).
female(mary).
female(susan).

% Rules
father(X, Y) :- parent(X, Y), male(X).
mother(X, Y) :- parent(X, Y), female(X).
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
siblings(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.

% Interactive query
run_family :-
    write('Enter relation (father/mother/grandparent/siblings): '), read(Rel),
    write('Enter first person: '), read(P1),
    write('Enter second person: '), read(P2),
    (call(Rel, P1, P2) -> writeln('True') ; writeln('False')).
```

OUTPUT:

A screenshot of the SWI-Prolog console window. The window title is "SWI-Prolog console". The menu bar includes "File", "Settings", "Tools", "Debug", and "Help". The console text shows the following sequence of events: 1. A welcome message: "Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2). SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software. Please run ?- license. for legal details." 2. A help message: "For online help and background, visit https://www.swi-prolog.org For built-in help, use ?- help(Topic). or ?- apropos(Word)." 3. Command 8: "?- consult(['c:/Users/kames/OneDrive/Documents/family-facts.pl'])." followed by the output "true." 4. Command 9: "?- run_family." followed by prompts "Enter relation (father/mother/grandparent/siblings): mother.", "Enter first person: |: john.", and "Enter second person: |: mike.". The output is "False" followed by "true." 5. Command 10: "?- run_family." followed by prompts "Enter relation (father/mother/grandparent/siblings): father.", "Enter first person: |: john.", and "Enter second person: |: mike.". The output is "True" followed by "true." 6. Command 10: "?- " followed by a red cursor.